

INTERNET OF THINGS

Assignment 1

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A study of different IoT Boards

1. Introduction to IoT Hardware

The Internet of Things (IoT) is a system of interconnected physical devices embedded with sensors, processors, and communication technologies that enable them to collect and exchange data over the internet.

An IoT system generally consists of:

- Sensors & Actuators
- IoT Development Board
- Connectivity (WiFi, Bluetooth, GSM, LoRa, etc.)
- Cloud Platform
- Application Layer (Dashboard / Mobile App)

Therefore, the IoT board acts as the brain of the system.

2. What is an IoT Board?

An IoT board is a microcontroller or microprocessor-based hardware platform used to:

- Read sensor data
- Process data
- Communicate over the internet
- Control actuators

IoT boards are mainly divided into:

TYPE	EXAMPLE	USE CASE
Microcontroller-based	Arduino, ESP32	Real-time, low power
Microprocessor-based	Raspberry Pi	Edge computing, AI

3. Case Study of Popular IoT Boards

3.1 Arduino Uno

Arduino Uno is one of the most beginner-friendly microcontroller boards used in embedded systems and IoT.

Technical Specifications:

- Microcontroller: ATmega328P
- Clock Speed: 16 MHz
- Operating Voltage: 5V
- Digital I/O Pins: 14
- Analog Pins: 6
- Connectivity: Requires external WiFi module (ESP8266)

Features:

- Easy programming (Arduino IDE)
- Large community support
- Low cost
- Ideal for prototypes

Limitations:

- No built-in WiFi
- Limited memory
- Not suitable for heavy data processing

Example Use Case:

Smart irrigation system with soil moisture sensor.

3.2 ESP8266 (NodeMCU)

ESP8266 is a low-cost WiFi-enabled microcontroller widely used in IoT applications.

Technical Specifications:

- 32-bit processor
- Built-in WiFi
- Operating Voltage: 3.3V
- GPIO pins available
- Flash memory: 4MB (varies)

Features:

- Built-in TCP/IP stack
- Low power consumption
- Cheap and compact

- Supports MQTT, HTTP

Advantages:

- Direct internet connectivity
- Ideal for cloud-based projects

Example:

Smart home light control mobile app.

3.3 ESP32

ESP32 is an advanced version of ESP8266 with WiFi + Bluetooth

Technical Specification:

- Dual-core processor
- WiFi + Bluetooth (BLE)
- Multiple ADCs, DACs
- Deep sleep mode

Advanced Features:

- Edge processing capability
- OTA (Over-the-Air updates)
- Secure boot & encryption

Applications:

- Wearables
 - Smart security systems
 - AI-based IoT devices
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3.4 Raspberry Pi

Raspberry Pi is a single-board computer (SBC) capable of running a full operating system like Linux.

Specifications (Pi 4):

- Quad-core processor
- 2GB/4GB/8GB RAM
- Built-in WiFi & Bluetooth
- HDMI support
- USB ports

Key Differences:

Unlike Arduino, Raspberry Pi runs an OS and supports:

- Python
- AI/ML models
- Computer vision
- Database servers

Example:

Smart surveillance system with face recognition.

4. Comparison of IoT Boards

Feature	Arduino	ESP8266	ESP32	Raspberry Pi
Type	Microcontroller	Microcontroller	Microcontroller	Microcontroller
Processing Power	Low	Medium	High	Very High
Ideal For	Beginners	Basic IoT	Advanced IoT	Edge AI
Built-in WiFi	✗	✓	✓	✓
Bluetooth	✗	✗	✓	✓
OS Support	✗	✗	✗	✓

5. Microcontroller vs Microprocessor

Microcontroller	Microprocessor
Used in Arduino, ESP	Used in Raspberry Pi
No OS	Runs Linux
Low power	Higher Power
Real-time systems	Complex computing

6. Power Consumption Analysis

- Arduino: Low
- ESP8266: Low
- ESP32: Very low (deep sleep)

- Raspberry Pi: High
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7. Security Features in Modern IoT Boards

Advanced boards like ESP32 support:

- Secure Boot
- Flash Encryption
- SSL/TLS
- OTA firmware updates

Security is critical because IoT devices are internet-facing.

8. Conclusion:

- Arduino is best for beginners
 - ESP8266 is best for low-cost internal projects
 - ESP32 is best for advanced IoT
 - Raspberry Pi is best for AI and Edge computing
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